

Question 9: In case of well-defined utility function, show that

$$\text{Concavity} \xrightarrow{\quad} \text{Quasi-concavity}$$

Answer:

In case of well-defined utility function, concavity implies quasi-concavity but the converse relation is not necessarily true.

$$\text{Concavity} \xrightarrow{\quad} \text{Quasi-concavity}$$

If a function $f(\cdot)$ is concave, then for any two points $x, y \in X$,

$$\begin{aligned} f(\alpha x + (1 - \alpha)y) &\geq \alpha f(x) + (1 - \alpha)f(y) \\ &\geq \min\{f(x), f(y)\} \end{aligned}$$

for all $\alpha \in (0,1)$.

where the first inequality follows from the definition of concavity, and the second holds true for all concave functions, as depicted in next figure. From the first and third element in the above inequality, we obtain that $f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\}$, which coincides with the definition of quasi-concavity. As a consequence, concavity implies quasi-concavity.

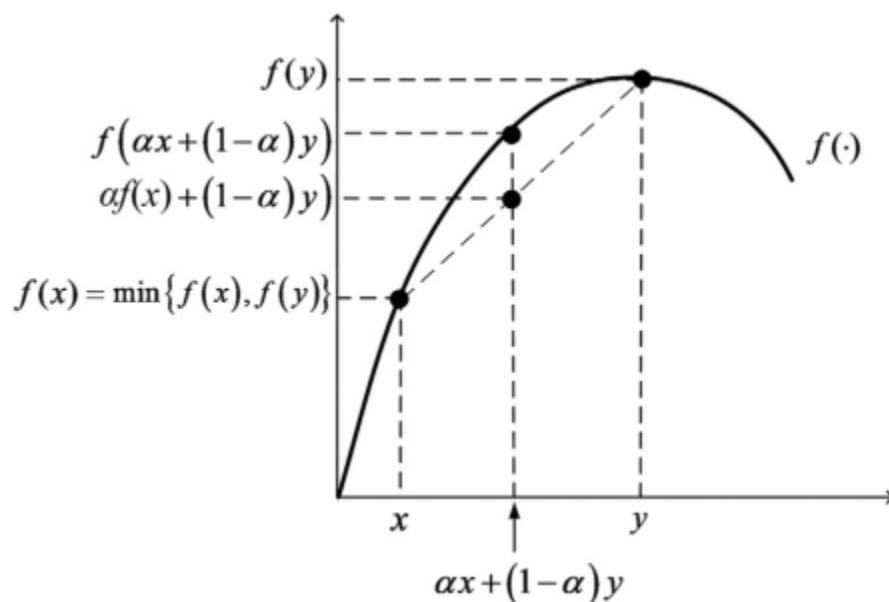


Figure: Concavity implies quasi-concavity

A concave $u(\cdot)$ exhibits diminishing marginal utility. That is, for an increase in the consumption bundle, the increase in utility is smaller as we move away from the origin.

The “jump” from one indifference curve to another requires:

- a slight increase in the amount of x_1 and x_2 when we are close to the origin
- a large increase in the amount of x_1 and x_2 as we get further away from the origin

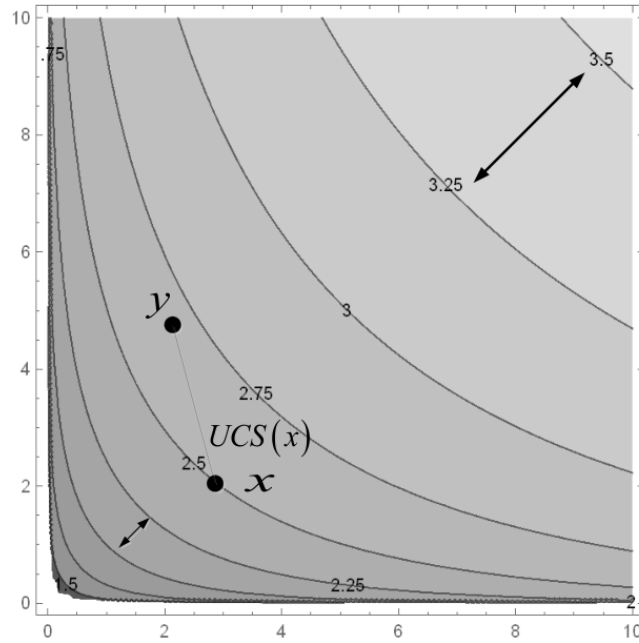


Fig: Concave and quasi-concave utility function (2D)

A convex $u(\cdot)$ exhibits increasing marginal utility. That is, for an increase in the consumption bundle, the increase in utility is larger as we move away from the origin.

The “jump” from one indifference curve to another requires:

- a large increase in the amount of x_1 and x_2 when we are close to the origin
- a small increase in the amount of x_1 and x_2 as we get further away from the origin

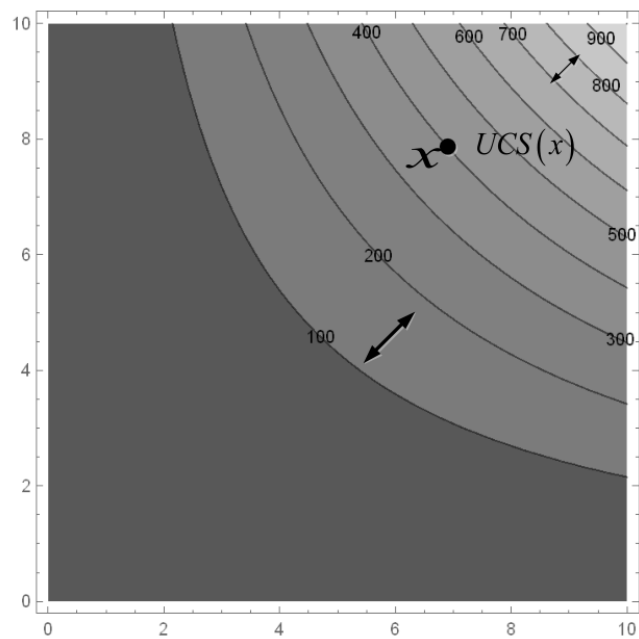


Fig: Convex but quasi-concave utility function (2D)

For any bundle x , the UCS(x) defines a convex set, and therefore the utility function satisfies the definition of quasi-concavity. Therefore, quasi-concavity (convex UCS) does not necessarily imply the concavity of a utility function.

In case of well-defined utility function, concavity implies quasi-concavity but quasi-concavity does not imply concavity.

Question 10: What are the typical utility functions used in Economic applications. Write down their functional form and calculate and interpret the notion of MRS.

Answer:

Here are some of the typical utility functions used in economic applications and their functional form:

- i. Cobb-Douglas Utility Functions
- ii. Perfect Substitutes
- iii. Perfect Complements
- iv. CES (constant elasticity of substitution) Utility Function
- v. Quasi-linear Utility Function

i. Cobb-Douglas Utility Functions: In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = Ax_1^\alpha x_2^\beta$$

where $A, \alpha, \beta > 0$.

Applying logs on both sides, we get:

$$\log u = \log A + \alpha \log x_1 + \beta \log x_2$$

Hence, the exponents in the original $u(\cdot)$ can be interpreted as elasticities:

$$\epsilon_{u, x_1} = \frac{\partial u(x_1, x_2)}{\partial x_1} \cdot \frac{x_1}{u(x_1, x_2)} = \alpha Ax_1^{\alpha-1} x_2^\beta \cdot \frac{x_1}{Ax_1^\alpha x_2^\beta} = \alpha$$

Intuitively, a one-percent increase in the amount of good x_1 increases individual utility by α percent. A similar interpretation applies to $\epsilon_{u, x_2} = \beta$.

Special cases are those in which

- $\alpha + \beta = 1$, such as $u(x_1, x_2) = Ax_1^\alpha x_2^{1-\alpha}$
- $A = 1$, such as $u(x_1, x_2) = x_1^\alpha x_2^\beta$
- $A = \alpha = \beta = 1$, such as $u(x_1, x_2) = x_1 x_2$

Besides yielding positive marginal utilities for all goods, whereby $\frac{\partial u}{\partial x_1} > 0$ and $\frac{\partial u}{\partial x_2} > 0$, and thus satisfying strong monotonicity, the Cobb-Douglas utility function yields a diminishing MRS, since

$$MRS_{x_1, x_2} = \frac{\alpha Ax_1^{\alpha-1} x_2^\beta}{\beta Ax_1^\alpha x_2^{\beta-1}} = \frac{\alpha x_2}{\beta x_1}$$

which is decreasing in x_1 . Hence, indifference curves become flatter as x_1 increases.

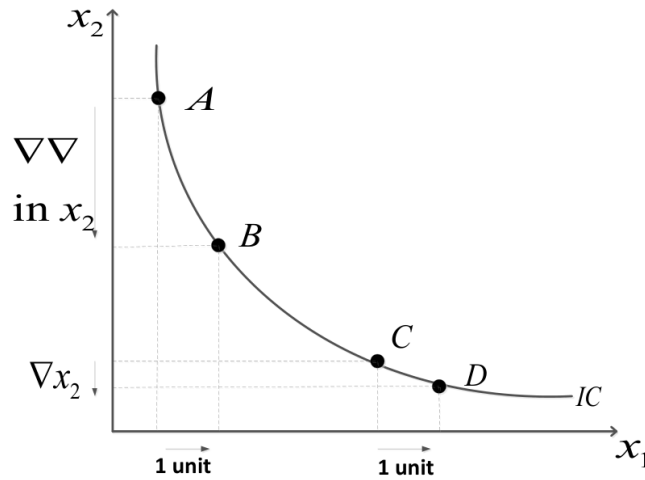


Fig: Cobb-Douglas preference

Individuals with Cobb–Douglas preferences are willing to give up many units of x_2 in order to increase their consumption of x_1 by one unit when they do not enjoy many units of x_1 (movement from point A to B in the figure), but such willingness to give up units of x_2 to obtain one more unit of x_1 decreases as x_1 becomes relatively abundant (movement from point C to D).

ii. **Perfect Substitutes:** In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = Ax_1 + Bx_2$$

where $A, \alpha, \beta > 0$.

As a consequence, the marginal utility of every good is constant, that is, $\frac{\partial u}{\partial x_1} = A$ for all x_1 and $\frac{\partial u}{\partial x_2} > B$ for all x_2 . The MRS is also constant, since $MRS_{x_1, x_2} = \frac{A}{B}$ does not depend on the amount of goods 1 or 2. Therefore indifference curves are straight lines with a slope of $-\frac{A}{B}$, as shown in figure.

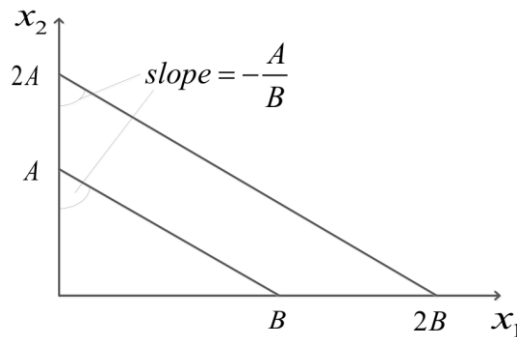


Fig: Perfect substitutes

Intuitively, the individual is willing to give up $\frac{A}{B}$ units of x_2 in order to obtain one more unit of x_1 and keep his utility level unaffected. Such willingness is, unlike in the Cobb–Douglas case, independent in the relative abundance of the two goods.

Examples: butter and margarine, coffee, and black tea, or two brands of unflavored mineral water.

iii. **Perfect Complements:** In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = A \cdot \min \{\alpha x_1, \beta x_2\}$$

where $A, \alpha, \beta > 0$.

Intuitively, increasing one of the goods without increasing the amount of the other good entails no increase in utility, which only goes up if the amounts of both goods increase (although they do not need to increase by the same amount).

The indifference curve is right angle with a kink at $\alpha x_1 = \beta x_2$, which, after rearranging, yields ray $x_2 = \frac{\alpha x_1}{\beta}$ from the origin, as depicted in figure.

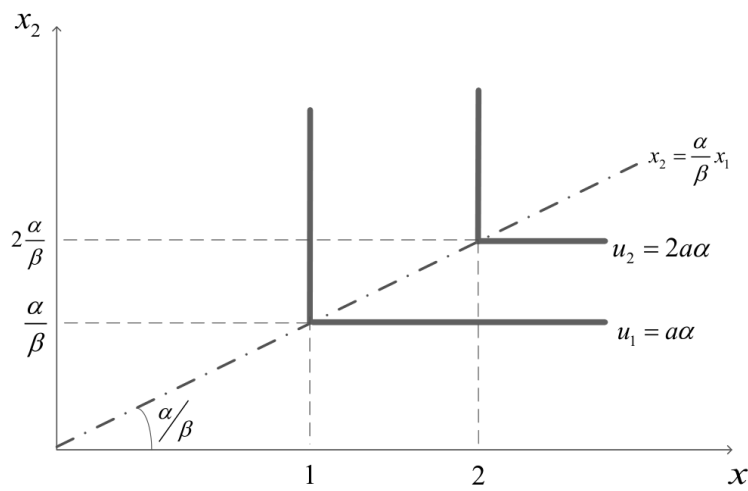


Fig: Perfect complements

The slope of such ray, $\frac{\alpha}{\beta}$, indicates the rate at which goods x_1 and x_2 must be consumed in order to achieve utility gains.

Special case: $\alpha = \beta$

$$\begin{aligned} u(x_1, x_2) &= A \cdot \min\{\alpha x_1, \alpha x_2\} \\ &= A\alpha \cdot \min\{x_1, x_2\} \\ &= B \cdot \min\{x_1, x_2\} \text{ if } B \equiv A\alpha \end{aligned}$$

Examples: cars and gasoline, or peanut butter and jelly. Other food recipes, which often require the use of ingredients in a fixed proportion, are good examples as well.

iv. **Constant elasticity of substitution (CES):** In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = \left[ax_1^{\sigma-\frac{1}{\sigma}} + bx_2^{\sigma-\frac{1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

where σ measures the elasticity of substitution between goods x_1 and x_2 .

In particular,

$$\sigma = \frac{\partial \left(\frac{x_2}{x_1} \right)}{\partial MRS_{1,2}} \frac{MRS_{1,2}}{\frac{x_2}{x_1}}$$

Here,

- $\sigma = 0$ under perfect complements (where substituting units of x_1 and x_2 without affecting the utility level is impossible),
- $\sigma = +\infty$ under perfect substitutes (where goods x_1 and x_2 can be exchanged at the same rate),
- $\sigma = 1$ under the Cobb–Douglas utility function.

Figure depicts the indifference curves of the CES utility function evaluated at different values of σ .

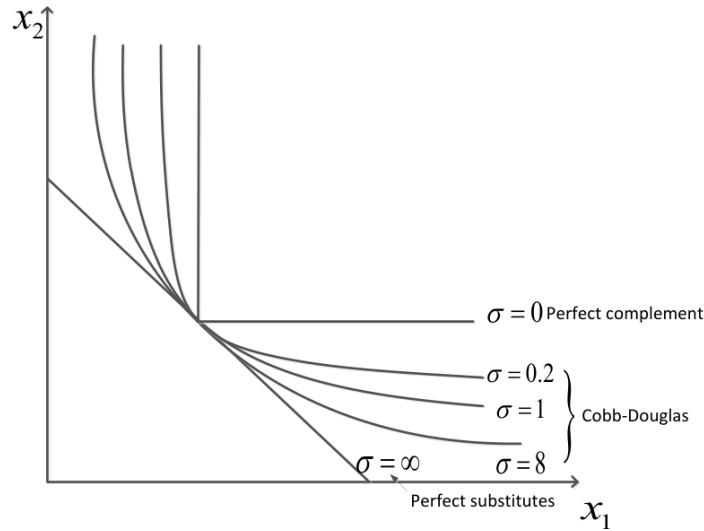


Fig: Constant elasticity of substitution (CES)

CES utility function is often presented as

$$u(x_1, x_2) = [ax_1^\rho + bx_2^\rho]^{\frac{1}{\rho}}$$

where, $\rho \equiv \frac{\sigma-1}{\sigma}$.

v. **Quasi-linear Utility Function:** In the case of two goods, x_1 and x_2 ,

$$u(x_1, x_2) = v(x_1) + bx_2$$

where x_2 enters linearly, $b > 0$, and $v(x_1)$ is a nonlinear function of x_1 . For example, $v(x_1) = a \ln x_1$ or $v(x_1) = ax_1^\alpha$, where $a > 0$ and $a \neq 1$.

The MRS is constant in the good that enters linearly in the utility function (x_2 in our case). In other words, for a given level of good 1, an increase in the amounts of good 2 (the desirable good that enters linearly) does not affect the slope of the indifference curve as depicted in figure.

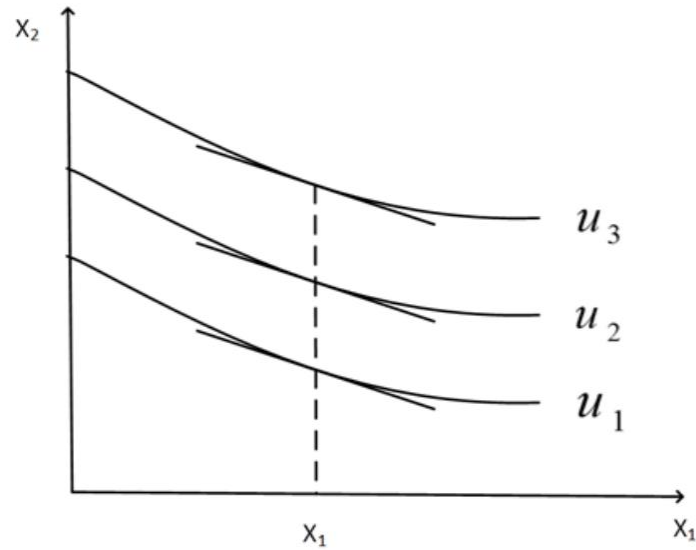


Fig: MRS of quasilinear preferences

Hence, at a given level of x_1 , a move to indifference curves farther away from the origin (larger amount of x_2) does not change its slope.

For the quasi-linear utility function $u(x_1, x_2) = v(x_1) + bx_2$, we have that $MU_{x_1} = \frac{\partial v}{\partial x_1}$ and $MU_{x_2} = b$, which implies that

$$MRS_{x_1, x_2} = \frac{\frac{\partial v}{\partial x_1}}{b}$$

which is constant in the good entering linearly, x_2 .

Quasilinear preferences are often used to represent the consumption of goods that are relatively insensitive to income.

Examples: garlic, toothpaste, etc.

Question 11: Examine the following properties of preference relations:

- a) **Homogeneity**
- b) **Homotheticity**

Answer:

Homogeneity: A utility function is homogeneous of degree k if varying the amounts of all goods by a common factor $\alpha > 0$ produces an increase in the utility level by α^k . That is, for the case of two goods, x_1 and x_2 ,

$$u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$$

where $\alpha > 0$. This allows for:

- $\alpha > 1$ in the case of a common increase
- $0 < \alpha < 1$ in the case of a common decrease

Properties of homogeneity: There are three properties of homogeneity.

- i. The first-order derivative of a function $u(x_1, x_2)$, which is homogeneous of degree k , is homogeneous of degree $k - 1$. In particular, by homogeneity of degree k , $u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$. Differentiating with respect to x_i for $i = \{1, 2\}$ yields

$$\begin{aligned} \frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_i} \cdot \alpha &= \alpha^k \cdot \frac{\partial u(x_1, x_2)}{\partial x_i} \\ \Rightarrow u'(\alpha x_1, \alpha x_2) &= \alpha^{k-1} u'(x_1, x_2) \end{aligned}$$

Therefore $u'(x_1, x_2)$ is homogeneous of degree $k - 1$.

- ii. The indifference curves of homogeneous functions are radial expansions of one another. That is, if two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve, i.e., $u(\alpha y) = u(\alpha z)$.
- iii. The MRS of a homogeneous function is constant for all points along each ray from the origin. That is, the slope of the indifference curve at point y coincides with the slope at a “scaled-up version” of point y , αy , where $\alpha > 1$. Since the MRS at bundle $x = x_1, x_2$ is

$$MRS_{1,2}(x_1, x_2) = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}}$$

the MRS at $(\alpha x_1, \alpha x_2)$ is

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\alpha^{k-1} \frac{\partial u(x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \frac{\partial u(x_1, x_2)}{\partial x_2}} = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}}$$

where the last equality uses the first property. Hence, the MRS is unaffected along all the points crossed by a ray from the origin.

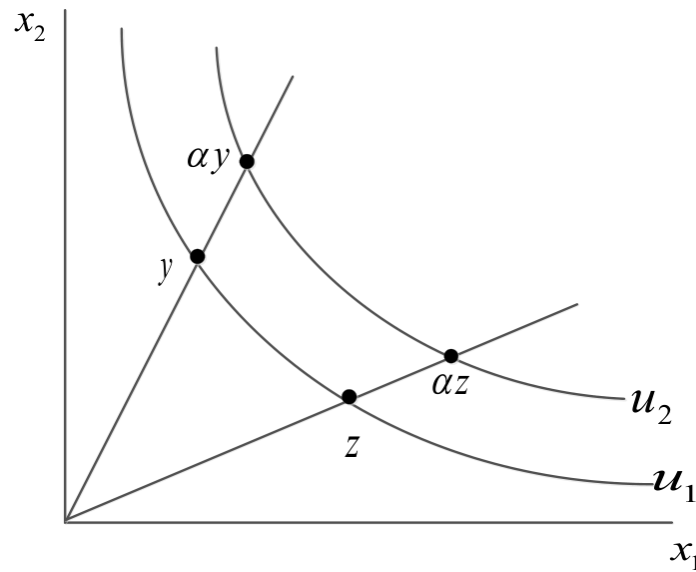


Fig: Homogeneous preference

Homotheticity: A utility function $u(x)$ is homothetic if it is a monotonic transformation of a homogeneous function. That is, $u(x) = g(v(x))$, where

- $g: \mathbb{R} \rightarrow \mathbb{R}$ is a strictly function, and
- $v: \mathbb{R}^n \rightarrow \mathbb{R}$ is homogeneous of degree k .

Properties of homotheticity: There are two properties of homotheticity.

- If preferences, $u(x)$, is homothetic, and two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve, i.e., $u(\alpha y) = u(\alpha z)$ for all $\alpha > 0$, as shown in figure.

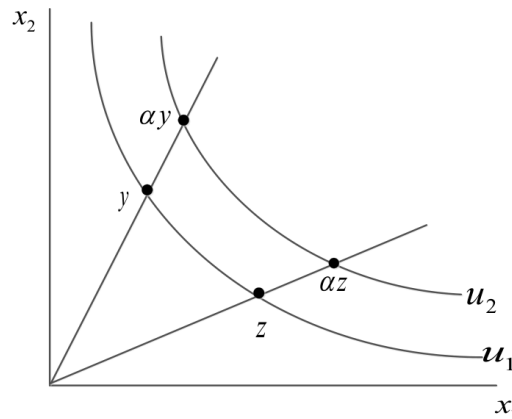


Fig: Homogeneous preference

Proof

If $u(y) = u(z)$ for two bundles y, z , then

$$u(\alpha y) = g(v(\alpha y)) = g(\alpha^k v(y))$$

where the first equality follows from homotheticity, and the second from the property that $v(x)$ is homogeneous of degree k . Since we can construct a similar equality for bundle z ,

$$u(\alpha z) = g(v(\alpha z)) = g(\alpha^k v(z))$$

and $v(y) = v(z)$, implying that $u(\alpha y) = u(\alpha z)$.

- ii. The MRS of a homothetic function is homogeneous of degree zero.

Proof

Let us first find $MRS_{1,2}(\alpha x_1, \alpha x_2)$,

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}}$$

where, $u(x_1, x_2) \equiv g(v(x_1, x_2))$.

Canceling the $\frac{\partial g}{\partial v}$ terms yields

$$\frac{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\alpha^{k-1} \cdot \frac{\partial v(x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \cdot \frac{\partial v(x_1, x_2)}{\partial x_2}}$$

Canceling the α^{k-1} terms yields

$$\frac{\frac{\partial v(x_1, x_2)}{\partial x_1}}{\frac{\partial v(x_1, x_2)}{\partial x_2}}$$

In summary,

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}} = \frac{\frac{\partial v(x_1, x_2)}{\partial x_1}}{\frac{\partial v(x_1, x_2)}{\partial x_2}} = MRS_{1,2}(x_1, x_2)$$